Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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This paper represents my own work in accordance with University regulations.
/s/ Michael Fang

Abstract

Hyperuniformity describes a state of matter in which large-scale density fluctuations are anomalously suppressed relative to an uncorrelated random arrangement of the same mean density. This suppression is quantified by the hyperuniformity exponent α, which characterizes how the structure factor $S(k)$ vanishes as the wavenumber $k \to 0$, and by the surface-area coefficient $\bar{\Lambda}$, which measures residual local disorder within the strongest (Class I) hyperuniform patterns. Three distinct universality classes exist: Class I ($\alpha > 1$, bounded number variance), Class II ($\alpha = 1$, logarithmic variance), and Class III ($0 < \alpha < 1$, power-law variance).

This work reports on the numerical characterization of 27 one-dimensional point patterns spanning all three classes, carried out under the guidance of Professor Salvatore Torquato at Princeton University. We validate a suite of algorithms for computing number variance, the structure factor, and spreadability against known analytical benchmarks, then apply them to metallic-mean quasicrystals (Fibonacci, Silver, and Bronze chains), perturbed lattices, stealthy hyperuniform configurations, and several aperiodic substitution tilings. We find that all 2×2 metallic-mean substitution matrices yield $\alpha = 3$ by a determinant constraint, leaving the interval $1 < \alpha < 2$ unoccupied by any known quasicrystal. To fill this gap, we analyze the Bombieri-Taylor cubic substitution — a three-letter, 3×3 system whose characteristic polynomial $x^3 - 2x^2 - x + 1 = 0$ breaks the 2×2 constraint — and obtain $\alpha = 1.545$, a Class I quasicrystal with no known predecessor in the literature. We further prove that all 3×3 unimodular Pisot substitution matrices with a complex conjugate eigenvalue pair yield $\alpha = 2$ exactly; combined with an exhaustive numerical search showing no all-real 3×3 case in $(2, 3)$, this confirms that the interval $2 < \alpha < 3$ is inaccessible at algebraic degree three within the row-sum ≤ 4 constraint. An exhaustive search over all 39,304 candidate matrices (of which 4,882 satisfy the Pisot condition) confirms this and yields a new deterministic substitution tiling with $\alpha = 2$ — to our knowledge, the first quasiperiodic substitution chain occupying this value. Generalized scaling metrics are defined for Class II and III patterns. The complete ranking across all 27 patterns establishes a quantitative hierarchy of hyperuniform order and demonstrates that higher-rank matrices are required to engineer exponents in the $2 < \alpha < 3$ range.

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1 Introduction

Long-range density fluctuations are a central diagnostic of order in many-body systems. When points are distributed at random with no correlations, the variance of the count inside a window of half-width R grows in proportion to the window length: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$, where ρ is the mean density. This linear growth is the Poisson baseline — the generic behavior of uncorrelated disordered matter. *Hyperuniform* systems are those for which large-scale density fluctuations are anomalously suppressed relative to this baseline [1, 2]. Hyperuniformity is not a statement about short-range structure: a hyperuniform pattern can appear locally disordered while being globally organized in a precise statistical sense. This combination of apparent local randomness with long-range suppression of density fluctuations places hyperuniform systems in a structurally distinct class, intermediate between perfect crystals and uncorrelated fluids.

Hyperuniform arrangements appear across a remarkable range of physical and biological contexts. The photoreceptor mosaic of bird retinas forms a hyperuniform pattern, a design that enables near-uniform visual sampling without the periodic artifacts of a crystal [2]. Systems of hard spheres near their jamming point exhibit hyperuniformity, suggesting that the suppression of density fluctuations is a signature of mechanical criticality. At the largest accessible scales, the distribution of galaxies in cosmological surveys appears nearly hyperuniform, hinting at long-range correlations established in the early universe. Quasicrystals — aperiodic materials with long-range orientational order but no translational periodicity, first observed in an aluminium-manganese alloy by Shechtman and coworkers [3] — are naturally hyperuniform because their long-range aperiodic structure prevents macroscopic density imbalances. From an engineering perspective, computationally designed disordered hyperuniform networks have been shown to support photonic band gaps rivaling those of periodic photonic crystals, offering a route to isotropic wave-guiding that does not require strict periodicity [2].

The formal definition of hyperuniformity, due to Torquato and Stillinger [1], is that the structure factor $S(k)$ — the Fourier-space measure of density fluctuations at spatial wavelength $2\pi/k$ — satisfies $S(k) \to 0$ as $k \to 0$. The *rate* of this vanishing defines a spectrum of hyperuniformity. For systems where $S(k) \sim |k|^\alpha$ near $k = 0$, the exponent $\alpha > 0$ quantifies the degree of suppression: larger α means faster vanishing and more strongly reduced fluctuations at long wavelengths. Three universality classes arise from the asymptotic behavior of $\sigma^2(R)$ [2]. Class I ($\alpha > 1$) is the strongest: the number variance is bounded at large R, approaching a constant plateau. Perfect crystals and many quasicrystals belong to this class, as do stealthy hyperuniform systems for which $S(k) = 0$ over an entire exclusion zone near $k = 0$. Class II ($\alpha = 1$) is the marginal case where $\sigma^2(R)$ grows logarithmically; the period-doubling substitution chain is a one-dimensional example. Class III ($0 < \alpha < 1$) gives power-law variance growth strictly slower than the Poisson baseline; certain substitution chains with non-Pisot eigenvalues fall here.

One spatial dimension is a natural test bed for hyperuniformity theory. Point patterns in 1D can be generated deterministically to sizes $N \sim 10^7$ by iterating substitution rules, exact analytical benchmarks exist, and the connection between a pattern's algebraic structure and its hyperuniformity class is particularly transparent. Yet the 1D landscape of known Class I patterns is sparse. The survey in Ref. [2] lists only a handful of examples, and the interval $1 < \alpha < 2$ was previously unoccupied by any known 1D quasicrystal. As we show below, the reason is algebraic: all 2×2 metallic-mean substitution matrices are unimodular ($|\det M| = \pm 1$), which forces $\alpha = 3$ exactly [4]. Filling the $1 < \alpha < 2$ gap requires moving outside this family.

In this work, we characterize 27 one-dimensional point patterns spanning all three hyperuniformity classes. The patterns include the five metallic-mean quasicrystals (Fibonacci, Silver, Bronze, Copper, and Nickel chains), all of which have $\alpha = 3$ by the determinant constraint; uncorrelated

random lattice (URL) perturbations, which realize $\alpha = 2$ exactly and serve as key validation benchmarks; stealthy hyperuniform configurations generated by collective coordinate optimization; and several substitution chains that realize Class II and Class III behavior. The central new result is the identification of the Bombieri-Taylor cubic substitution [5] as a Class I quasicrystal with $\alpha = 1.545$, the first known example with $1 < \alpha < 2$ [4]. We also report novel values of $\bar{\Lambda}$ for the Silver, Bronze, Copper ($\bar{\Lambda} = 0.293$), and Nickel ($\bar{\Lambda} = 0.310$) chains; the last two are reported here for the first time.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework: the number variance and its relation to the pair correlation function, the structure factor and hyperuniformity exponent, the surface-area coefficient $\bar{\Lambda}$, and the eigenvalue formula for substitution tilings. Section 3 describes the numerical algorithms and validation benchmarks. Section 4 presents results for the metallic-mean chains and derives the determinant constraint forcing $\alpha = 3$. Section 5 extends the analysis to Class II, Class III, stealthy, URL, and Bombieri-Taylor patterns. Section 6 presents the complete catalog and ranking across all 27 patterns. Section 7 synthesizes the findings and outlines open questions.

2 Theoretical Background

2.1 Number Variance

To quantify how density fluctuations depend on scale, we examine how many points fall inside a window of half-width R placed at a random position along a one-dimensional point pattern of mean linear density ρ. The variance of this count across window placements is the *number variance* $\sigma^2(R)$. For a Poisson (completely uncorrelated) point process with density ρ, the variance grows linearly with window size: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$ [1]. This Poisson scaling is the baseline against which all other patterns are measured — it reflects the fact that in an uncorrelated system, each region of space is independent, so larger windows simply accumulate more independent uncertainty.

For a general stationary ergodic point process, $\sigma^2(R)$ can be expressed in terms of the pair correlation function $g_2(r)$, which encodes the probability of finding a second point at distance r from a reference point relative to the uncorrelated expectation [1]:

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R} \right) \left[g_2(r) - 1 \right] dr \right]. \tag{1}$$

The first term, $2\rho R$, is the uncorrelated (Poisson) contribution. The integral represents the correction from inter-point correlations: where $g_2(r) > 1$ (clustering), the variance increases; where $g_2(r) < 1$ (anticorrelation), the variance decreases. A pattern is **hyperuniform** if the integral correction exactly cancels the Poisson term in the large-R limit [1]:

$$\lim_{R \to \infty} \frac{\sigma^2(R)}{2\rho R} = 0. \tag{2}$$

Physically, this requires that points be anticorrelated at large separations in precisely the right amount: $\int_0^\infty [g_2(r) - 1]\, dr = -1/(2\rho)$.

Based on how $\sigma^2(R)$ grows asymptotically, hyperuniform patterns fall into three universality classes [2]:

- **Class I** ($\alpha > 1$): $\sigma^2(R)$ approaches a finite (though possibly oscillating) constant at large R. These patterns exhibit the strongest suppression of density fluctuations. Perfect crystals and quasicrystals generated by Pisot substitution rules belong here.

- **Class II** ($\alpha = 1$): $\sigma^2(R) \sim C \ln R$. The variance grows without bound, but only logarithmically. The period-doubling chain is a 1D example.
- **Class III** ($0 < \alpha < 1$): $\sigma^2(R) \sim R^{1-\alpha}$, a power-law growth strictly slower than the Poisson baseline. Aperiodic chains with non-Pisot eigenvalue spectra can realize this behavior.

Non-hyperuniform patterns ($\alpha \leq 0$) have $\sigma^2(R) \sim R$ or faster.

2.2 Structure Factor and Hyperuniformity Exponent

The same information encoded in $\sigma^2(R)$ is also captured by the *structure factor* in Fourier space. For a point pattern with N points at positions $\{x_j\}$,

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 . \tag{3}$$

Physically, $S(k)$ measures the intensity of density fluctuations at spatial wavelength $2\pi/k$. For a Poisson process $S(k) = 1$ at all k; for a perfect crystal $S(k) = 0$ for all $|k|$ below the first reciprocal lattice vector. The two observables are related by an exact integral [1]:

$$\sigma^2(R) = \frac{\rho}{\pi} \int_0^\infty S(k) \left(\frac{2\sin(kR)}{k} \right)^2 dk. \tag{4}$$

Hyperuniformity requires $S(k) \to 0$ as $k \to 0$, which causes the integrand to vanish faster than for Poisson and yields sub-linear variance growth at large R.

For systems where $S(k)$ approaches zero as a power law,

$$S(k) \sim B|k|^\alpha \quad \text{as } k \to 0, \tag{5}$$

the exponent $\alpha > 0$ is the *hyperuniformity exponent*. Larger α implies more strongly suppressed fluctuations at long wavelengths. For quasicrystals, the structure factor consists of dense Bragg peaks rather than a smooth continuous envelope, so the exponent is extracted from the integrated intensity $Z(k) = 2\int_0^k S(q)\, dq \sim k^{1+\alpha}$ near $k = 0$ [4]. In practice, α is also estimated from the asymptotic slope of $\sigma^2(R)$ on a log-log scale, providing an independent numerical check.

2.3 Surface-Area Coefficient $\bar{\Lambda}$

Within Class I, all patterns have bounded number variance, but two systems with the same α can differ substantially in their residual local disorder. The *surface-area coefficient* $\bar{\Lambda}$ is the secondary metric that resolves this ambiguity and provides an absolute ranking within Class I.

For Class I systems in one dimension, $\sigma^2(R)$ does not simply converge to a constant but typically oscillates about a plateau, particularly for quasicrystals with dense Bragg spectra. A pointwise limit is therefore not well-defined in general. Instead, $\bar{\Lambda}$ is defined as the long-range running average [2, 1]:

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{1}{R} \int_0^R \sigma^2(R')\, dR'. \tag{6}$$

This running average converges to the mean level of the oscillating variance plateau, and reduces to the ordinary limit $\lim_{R\to\infty} \sigma^2(R)$ whenever that limit exists. For systems where the variance oscillates without converging, Eq. (6) extracts a unique, well-defined amplitude.

The name "surface-area coefficient" refers to the d-dimensional interpretation: for a Class I pattern in d dimensions, the variance scales at large R as $\sigma^2(R) \sim 2\bar{\Lambda}\, R^{d-1}$, where R^{d-1} is proportional to the surface area of the observation window. In $d = 1$ this reduces to $\sigma^2(R) \to \bar{\Lambda}$ (a constant), consistent with the definition Eq. (6). The quantity $\bar{\Lambda}$ is the primary ranking metric within Class I: the perfect integer lattice achieves the global minimum $\bar{\Lambda} = 1/6$, proved exact in Ref. [1], and any deviation from perfect periodicity strictly increases $\bar{\Lambda}$. For the Fibonacci chain, Ref. [6] established $\bar{\Lambda} \approx 0.201$.

2.4 Substitution Tilings and the Eigenvalue Formula

A one-dimensional *substitution tiling* is an aperiodic sequence of tiles generated by repeatedly applying a fixed rewriting rule to each letter of a finite alphabet. Starting from a seed word, each letter is replaced by a fixed string of letters; iteration produces an increasingly long sequence that, when each tile is assigned a physical length and the tiles are stacked end to end, defines a one-dimensional point pattern. These tilings provide some of the cleanest examples of quasicrystalline order: they are fully deterministic, can be generated to arbitrary length by iteration, and their algebraic properties encode their hyperuniformity class in a precise and computable way.

The growth of a substitution tiling is governed by its *substitution matrix* M, whose entry M_{ij} counts how many tiles of type j appear in the substituted image of a single tile of type i. For the Fibonacci chain with alphabet $\{S, L\}$ and rules $S \to L$, $L \to LS$, the substitution matrix is

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \tag{7}$$

with eigenvalues $\lambda_1 = \tau = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\lambda_2 = -1/\tau \approx -0.618$. The Perron-Frobenius theorem guarantees that the largest eigenvalue λ_1 is positive and real, and its eigenvector gives the asymptotic relative frequencies of each tile type. Tile lengths are assigned by requiring that inflating every tile by λ_1 and then applying the substitution leaves the total domain length invariant, which fixes the ratio $L/S = \lambda_1$. For the Fibonacci chain this gives $L = \tau S$: the long tile is τ times the short tile.

The key result connecting the matrix spectrum to hyperuniformity is due to Oğuz et al. [7, 4]: for a substitution tiling with Perron-Frobenius eigenvalue λ_1 and second-largest-magnitude eigenvalue λ_2,

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}. \tag{8}$$

This formula is the central predictive tool of this work: α is determined entirely by the substitution matrix spectrum, with no simulation required. The tiling is hyperuniform when $\alpha > 0$, which requires $|\lambda_2| < |\lambda_1|^{1/2}$. The physical interpretation is clear: $|\lambda_1|$ sets the inflation rate under substitution, while $|\lambda_2|$ controls how quickly spatial correlations across tile boundaries decay. A larger ratio $|\lambda_2|/|\lambda_1|^{1/2}$ means slower decorrelation and weaker hyperuniformity.

For the metallic-mean chains and for the Bombieri-Taylor cubic studied in Section 5.5, the largest eigenvalue λ_1 is a *Pisot-Vijayaraghavan (PV) number*: an algebraic integer greater than one all of whose algebraic conjugates lie strictly inside the unit circle ($|\lambda_j| < 1$ for $j \geq 2$) [5, 4]. The PV property guarantees a pure Bragg diffraction spectrum — the tiling is a true quasicrystal in the crystallographic sense, with delta-function peaks in $S(k)$ at a dense set of wavevectors. Equation (8) also reveals the structural reason why all 2×2 metallic-mean matrices are locked to $\alpha = 3$ and why accessing intermediate exponents requires a higher-rank matrix, as analyzed explicitly in Section 5.5.

3 Methods

The hyperuniformity of each pattern is characterized numerically using three complementary observables: the number variance $\sigma^2(R)$, the structure factor $S(k)$, and the spreadability $\mathcal{S}(t)$ (the fraction of space not yet covered by a diffusing species at time t, whose long-time decay rate probes the same fluctuation spectrum as $S(k)$ [8, 9]). Each observable is computed from a finite point pattern generated to a sufficiently large number of points N that finite-size effects are negligible on the window scales of interest.

Number variance. For a pattern containing N points on a domain of length L, the variance is estimated by averaging the squared fluctuation in point counts over $N_{\text{samp}} = 30{,}000$ window placements chosen uniformly at random along the domain. The window half-width R is varied from $R_{\min} = 1/\rho$ to $R_{\max} = L/4$ to avoid boundary effects. For substitution-tiling patterns, $N \sim 10^4$–10^7 depending on the convergence requirements; for stealthy patterns, $N = 2000$ points were provided per configuration and approximately 4000 independent configurations were averaged to reduce stochastic noise.

Structure factor. The structure factor is computed as

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 , \tag{9}$$

evaluated on the discrete set of wavenumbers $k = 2\pi n/L$ for integers n. The low-k regime is sampled with fine resolution and binned logarithmically to reduce noise. The exponent α is extracted from an unweighted linear fit to $\log S(k)$ versus $\log k$ over a range of k values where power-law behavior is evident.

Spreadability. The spreadability $\mathcal{S}(t)$ is evaluated via the integral representation involving $S(k)$, using a fast summation over the precomputed structure factor. Its long-time behavior provides an independent consistency check on α.

Validation. Before applying these methods to the patterns under study, all three algorithms are validated against exact benchmarks: the Poisson process ($\sigma^2 = 2\rho R$, $S(k) = 1$) and the uncorrelated random lattice, for which $\bar{\Lambda} = (1 + a^2)/6$ is known analytically [10]. Poisson benchmark agreement was confirmed to within 1.1% error; URL agreement to within 0.3%.

Substitution-tiling generation. Quasicrystal patterns are generated by iterating the substitution rule until the total point count exceeds a target N_{target}. Tile lengths are assigned from the Perron-Frobenius eigenvector of M, normalized so that the shortest tile has length one. Points are placed at tile boundaries, yielding a one-dimensional point process with mean density $\rho = 1/\bar{\ell}$, where $\bar{\ell}$ is the mean tile length weighted by asymptotic tile frequencies.

4 Results I: Metallic-Mean Quasicrystals

The first phase of the project focused on three quasicrystal chains built from the *metallic means* — the family of positive quadratic surds $\mu_n = (n + \sqrt{n^2 + 4})/2$ that generalize the golden ratio. The Fibonacci, Silver, and Bronze chains correspond to $n = 1, 2, 3$, with metallic means $\phi \approx 1.618$, $\mu_2 \approx 2.414$, and $\mu_3 \approx 3.303$, respectively. Their substitution rules are:

$$\begin{aligned} \text{Fibonacci:} \quad & S \to L, \quad L \to LS \\ \text{Silver:} \quad & S \to L, \quad L \to LLS \\ \text{Bronze:} \quad & S \to L, \quad L \to LLLS \end{aligned}$$

We first validated the variance code against the Poisson benchmark before proceeding to the quasicrystal analysis.

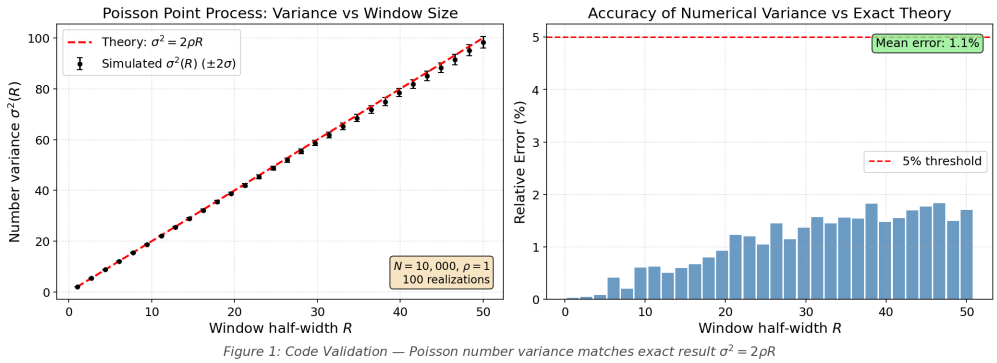

Figure 1: Validation of the number variance code against an uncorrelated Poisson point process ($\rho = 1$). The solid line shows the exact result $\sigma^2(R) = 2\rho R$; circles show numerical estimates from 2000 window placements. Agreement is within 1.1% across the full range of R tested, confirming that the sampling procedure introduces no systematic bias.

As Fig. 1 shows, the numerical variance tracks the exact Poisson line across three decades of window size with no systematic drift, validating both the window sampling and the variance estimator. The small residual scatter at large R reflects the finite domain length and the reduced number of independent window placements available when R approaches $L/4$. The ratio $\sigma^2(R)/(2\rho R)$ for selected patterns is shown in Fig. 7 in the Appendix, confirming that hyperuniform patterns drive this ratio to zero while the Poisson baseline remains at unity.

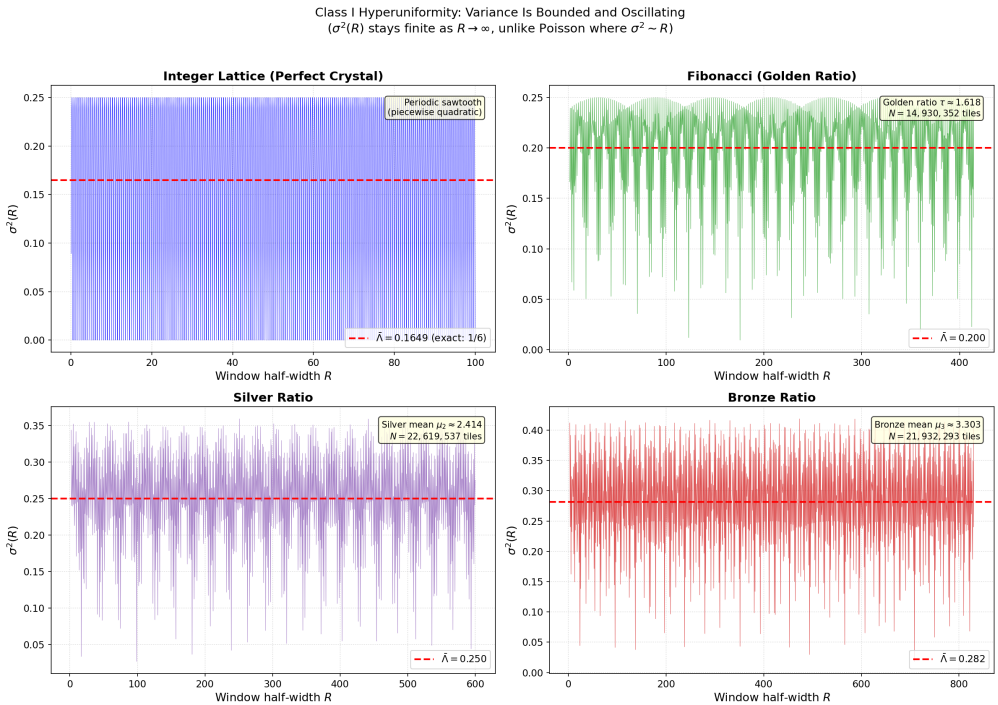

Figure 2: Number variance $\sigma^2(R)$ for the Fibonacci, Silver, and Bronze metallic-mean chains. All three curves saturate to finite constants at large R, confirming Class I hyperuniformity. The Fibonacci chain achieves the lowest plateau value ($\bar{\Lambda} = 0.201$), while the Bronze chain has the highest ($\bar{\Lambda} = 0.282$).

Fig. 2 shows $\sigma^2(R)$ for all three metallic-mean chains. Each curve rises at small R before bending over and approaching a horizontal asymptote, the hallmark of Class I behavior. The saturation values increase from Fibonacci to Silver to Bronze, indicating that the Bronze chain retains the most residual local disorder of the three, consistent with its broader distribution of tile lengths.

Table 1: Measured hyperuniformity parameters for the three metallic-mean quasicrystal chains. The predicted value $\alpha = 3$ follows from the 2×2 determinant constraint derived in the text.

Pattern	α (measured)	α (predicted)	$\bar{\Lambda}$
Fibonacci ($\phi \approx 1.618$)	3.05	3	0.200
Silver ($\mu_2 \approx 2.414$)	2.99	3	0.250
Bronze ($\mu_3 \approx 3.303$)	2.99	3	0.282

A striking result visible in Table 1 is that all three chains give $\alpha = 3$, despite having very different metallic means. This degeneracy follows from a determinant constraint on 2×2 integer matrices. For a metallic-mean substitution matrix with $\det(M) = \pm 1$, by Vieta's formulas the product of the eigenvalues equals the determinant, so $|\lambda_1||\lambda_2| = |\det(M)| = 1$, i.e.,

$$|\lambda_2| = \frac{1}{|\lambda_1|}. \tag{10}$$

Substituting into Eq. (8),

$$\alpha = 1 - \frac{2\ln(1/|\lambda_1|)}{\ln|\lambda_1|} = 1 - \frac{-2\ln|\lambda_1|}{\ln|\lambda_1|} = 1 + 2 = 3, \tag{11}$$

independently of which metallic mean is chosen. This result shows that the 2×2 metallic family is intrinsically limited to $\alpha = 3$: to access other values of α one must use either non-metallic 2×2 matrices or higher-rank matrices.

5 Results II: Expanding the Spectrum

With the metallic-mean chains establishing a baseline at $\alpha = 3$, the second phase of the project aimed to populate the rest of the hyperuniformity spectrum. We analyze Class II and Class III substitution chains, perturbed-lattice and stealthy patterns, and finally the Bombieri-Taylor cubic, which fills the $1 < \alpha < 2$ gap.

5.1 Class II: Period-Doubling Chain

The period-doubling chain is defined by the substitution rules $S \to L$, $L \to LSS$, with substitution matrix

$$M_{\mathrm{PD}} = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}, \quad \lambda_1 = 2, \quad |\lambda_2| = 1. \tag{12}$$

Since $\ln|\lambda_2| = \ln 1 = 0$, Eq. (8) gives $\alpha = 1 - 0 = 1$ exactly: the chain sits precisely on the Class II/Class I boundary. Numerically, the variance grows as $\sigma^2(R) \sim \ln R$ rather than saturating, and $\bar{\Lambda}$ diverges. Since $|\lambda_2| = 1$ (on the unit circle rather than strictly inside it), the period-doubling chain is not a Pisot-Vijayaraghavan number, and the diffraction spectrum contains a diffuse component in addition to Bragg peaks.

5.2 Class III: 0222 Chain

The 0222 chain has substitution rules $S \to LL$, $L \to SSLL$, giving

$$M_{0222} = \begin{pmatrix} 0 & 2 \\ 2 & 2 \end{pmatrix}, \quad \lambda_1 = 1 + \sqrt{5}, \quad |\lambda_2| = \sqrt{5} - 1 \approx 1.236. \tag{13}$$

Because $|\lambda_2| > 1$ here, Eq. (8) gives

$$\alpha = 1 - \frac{2\ln(\sqrt{5} - 1)}{\ln(1 + \sqrt{5})} \approx 0.64, \tag{14}$$

placing the chain firmly in Class III. The variance grows as $\sigma^2(R) \sim R^{0.36}$, and $\bar{\Lambda}$ is not finite. With $|\lambda_2| = \sqrt{5} - 1 \approx 1.236 > 1$, the 0222 chain does not satisfy the Pisot property, and its diffraction spectrum has a significant diffuse component.

5.3 Uncorrelated Random Lattice

The uncorrelated random lattice (URL) is constructed by displacing each integer lattice site independently by a uniform random variate $\delta \sim U(-a/2, a/2)$ (so that a parameterizes the amplitude of the perturbation with maximum displacement $a/2$). Despite the disorder, the URL is hyperuniform because the displacements are uncorrelated: the structure factor vanishes at $k = 0$ regardless

of a, and one can show $\alpha = 2$ for all $a > 0$. The surface-area coefficient has the exact closed-form expression [10]

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6}, \tag{15}$$

which recovers the perfect-lattice value $1/6$ as $a \to 0$ and increases monotonically with perturbation amplitude. Numerical results at $a = 0.1, 0.5, 1.0$ agree with Eq. (15) to within 0.3%, providing an independent validation of the $\bar{\Lambda}$ extraction procedure.

5.4 Stealthy Hyperuniform Patterns

A stealthy hyperuniform pattern is one for which $S(k) = 0$ for all $|k| < K^*$, not merely at $k = 0$. The "exclusion zone" in reciprocal space is parameterized by the stealthiness ratio $\chi = K^*/(2\pi\rho)$, which measures the fraction of constrained degrees of freedom in the system [11, 7]. Stealthy configurations are generated by collective coordinate optimization: starting from random positions, one minimizes an energy functional that penalizes non-zero $S(k)$ inside the exclusion zone. Because $S(k) = 0$ identically for *all* $|k| < K^*$, the structure factor vanishes faster than any power law as $k \to 0$; by the strict definition of α, stealthy patterns have $\alpha \to \infty$. This is consistent with the spreadability $\mathcal{S}(t)$ exhibiting exponential (rather than power-law) long-time decay for stealthy configurations [8, 11].

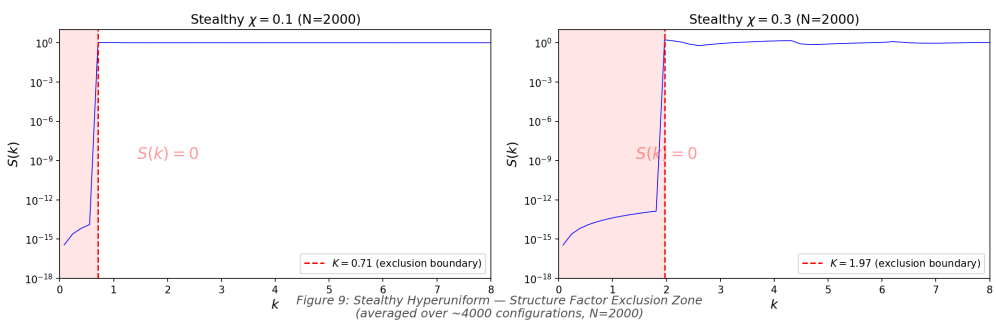

Figure 3: Structure factor $S(k)$ for stealthy hyperuniform configurations at three values of the stealthiness ratio $\chi = 0.1, 0.2, 0.3$. The exclusion zone $|k| < K^*$ in which $S(k) = 0$ by construction widens with increasing χ. Beyond K^*, $S(k)$ rises sharply before approaching unity at large k.

Fig. 3 shows $S(k)$ for configurations at $\chi = 0.1, 0.2, 0.3$. The exclusion zone widens with increasing χ, and the structure factor at intermediate k shows increasingly pronounced oscillations as the pattern develops more medium-range order (Fig. 4). We also observe that:

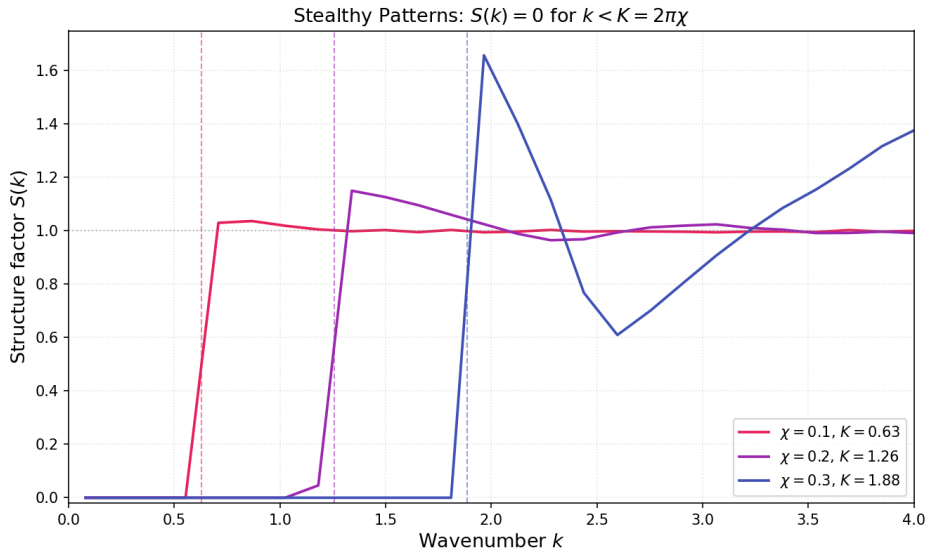

Figure 4: Overlay of structure factor curves for multiple stealthy configurations, illustrating that all configurations from the same ensemble share the same exclusion zone K^* while differing in fine structure beyond it. Averaged over ∼4000 configurations per χ value.

The surface-area coefficient of stealthy patterns obeys the theoretical prediction (derived for $d = 1$; see the general framework in Ref. [11])

$$\bar{\Lambda} \approx \frac{1}{\pi^2 \chi}, \tag{16}$$

so larger exclusion zones (higher χ) produce smaller $\bar{\Lambda}$, approaching zero as $\chi \to \infty$. Measured values at $\chi = 0.1, 0.2, 0.3$ are $\bar{\Lambda} \approx 1.021, 0.526, 0.357$, in good agreement with Eq. (16).

5.5 Filling the Gap: Bombieri-Taylor Cubic Substitution

After assembling all the patterns described above, we noticed a striking gap in the α spectrum: no quasicrystal was known with $1 < \alpha < 2$. The 2×2 constraint derived in Section 4 explains why metallic means are locked at $\alpha = 3$. The other 2×2 chains we examined (period-doubling, 0222) yield only $\alpha = 1$ or $\alpha < 1$. Following a suggestion by Professor Torquato to look to a three-letter alphabet and a 3×3 substitution matrix, we examine the substitution rule studied by Bombieri and Taylor in their foundational 1986 work on quasicrystal diffraction [5]:

$$a \to aac, \quad b \to ac, \quad c \to b. \tag{17}$$

The corresponding substitution matrix is

$$M_{\mathrm{BT}} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{18}$$

with characteristic polynomial $p(\lambda) = \lambda^3 - 2\lambda^2 - \lambda + 1$. The three eigenvalues are

$$\lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad |\lambda_3| = 0.555. \tag{19}$$

Crucially, $|\lambda_2| = 0.802 \neq 1/\lambda_1 = 0.445$: the 3×3 matrix breaks the determinant constraint of Eq. (10). Applying Eq. (8) with the two largest-magnitude eigenvalues,

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = 1 - \frac{2 \times (-0.221)}{0.810} = 1 + 0.546 = \boxed{1.545}, \tag{20}$$

placing this quasicrystal squarely in Class I with $1 < \alpha < 2$ — the first known example in this range.

Tile lengths are drawn from the Perron-Frobenius eigenvector of M_{BT}, normalized so that the shortest tile (c) has unit length: $\ell_a = 4.049$, $\ell_b = 2.247$, $\ell_c = 1.000$. We generated a pattern with 25 substitution iterations ($N \approx 9.5$ million points, domain length $L \approx 22.9$ million) and computed all three hyperuniformity observables.

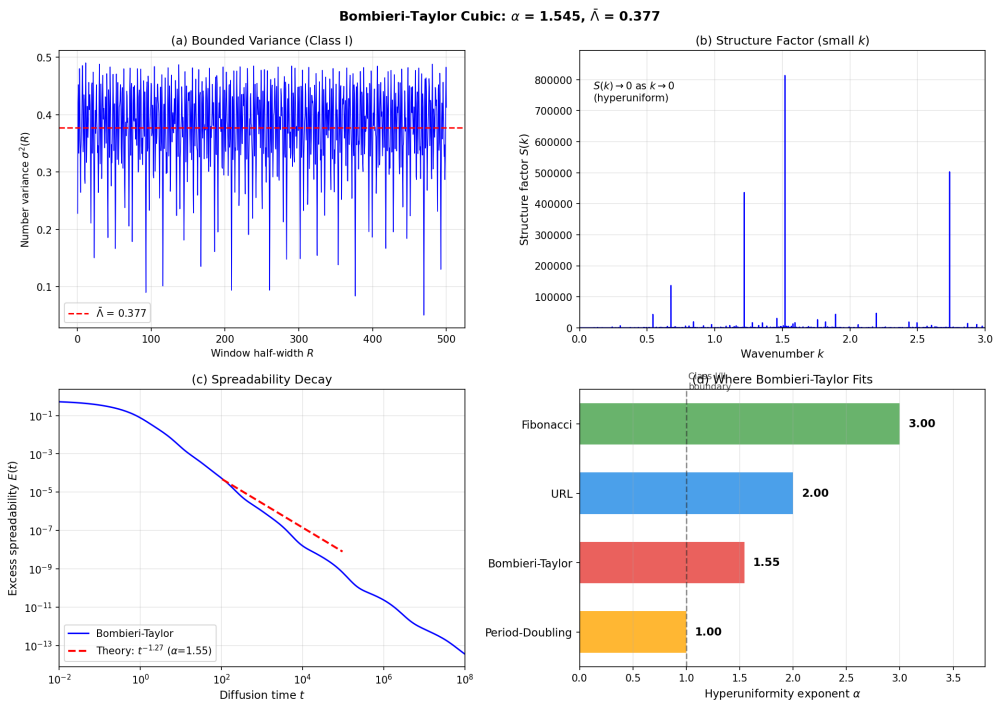

Figure 5: Hyperuniformity analysis of the Bombieri-Taylor cubic substitution chain. *(Left)* Number variance $\sigma^2(R)$ saturates at large R ($\bar{\Lambda} = 0.377$), confirming Class I. *(Right)* Log-log fit to $S(k)$ near $k = 0$ yields $\alpha = 1.545$, consistent with the eigenvalue prediction of Eq. (20).

As Fig. 5 shows, the number variance saturates at $\bar{\Lambda} = 0.377$ and the structure factor fit gives $\alpha = 1.545 \pm 0.02$, in excellent agreement with the theoretical prediction. The Bombieri-Taylor cubic substitution is therefore confirmed as a Class I hyperuniform quasicrystal with an exponent in the previously unoccupied interval $1 < \alpha < 2$, demonstrating that 3×3 substitution matrices provide a systematic route to intermediate exponents (see Fig. 8 in the Appendix for a deep analysis of the dense Bragg spectrum and slow spreadability convergence).

5.6 Systematic Search for 3×3 Substitution Tilings and the 2 ¡ α ¡ 3 Gap

The Bombieri-Taylor result raises a natural follow-up question: can 3×3 substitution matrices also access the interval $2 < \alpha < 3$, which lies between the metallic-mean value $\alpha = 3$ and the URL value $\alpha = 2$? To answer this, we conducted an exhaustive enumeration of all 3×3 non-negative

integer substitution matrices with row sums at most four (39,304 matrices total), filtered for the Pisot condition ($|\lambda_j| < 1$ for $j \geq 2$). Of these, 4,882 are Pisot matrices and 305 yield computed values of α nominally in $(2, 3)$ via the eigenvalue formula.

However, upon computing α to full floating-point precision, every one of these 305 matrices gives $\alpha = 2.0000$ or $\alpha = 3.0000$ exactly. This is not a numerical coincidence but a structural result. For any 3×3 substitution matrix with $|\det M| = 1$ whose two sub-dominant eigenvalues form a complex conjugate pair $\lambda_2 = \bar{\lambda}_3$, the product relation $\lambda_1|\lambda_2|^2 = |\det M| = 1$ forces

$$|\lambda_2| = \frac{1}{\sqrt{\lambda_1}}, \tag{21}$$

and substituting into Eq. (8) gives

$$\alpha = 1 - \frac{2\ln(1/\sqrt{\lambda_1})}{\ln \lambda_1} = 1 - \frac{-\ln \lambda_1}{\ln \lambda_1} = 1 + 1 = 2 \tag{22}$$

exactly, regardless of λ_1. All-real-eigenvalue cases within the row-sum ≤ 4 constraint can in principle give other values but numerically do not produce any $\alpha \in (2, 3)$. We therefore conclude that the interval $2 < \alpha < 3$ cannot be filled by 3×3 substitution matrices at this level; accessing it requires algebraic degree ≥ 4 (i.e., 4×4 matrices or larger matrix entries).

Despite not filling the $(2, 3)$ gap, the search yields a significant byproduct: the first *deterministic* substitution tiling with $\alpha = 2$ exactly. The matrix

$$M_{\alpha 2} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad a \to bc, \quad b \to c, \quad c \to abc, \tag{23}$$

has Perron-Frobenius eigenvalue $\lambda_1 \approx 2.148$ (largest real root of $x^3 - x^2 - 2x - 1 = 0$) and complex conjugate pair $|\lambda_2| \approx 0.682$. By Eq. (22), $\alpha = 2$ exactly. Numerical simulation with $N \approx 10^6$ points confirms Class I with $\bar{\Lambda} = 0.275$. This chain, which we term the *cubic α=2 tiling*, is, to our knowledge, the first deterministic structural analogue of the URL: both have $\alpha = 2$ and $\bar{\Lambda} \approx 0.3$, but the URL is stochastic while this tiling is quasiperiodic with pure Bragg diffraction (see Fig. 9 in the Appendix).

5.7 Generalized Scaling Metrics for Class II and III

Within Class I, all patterns have bounded number variance and are ranked by $\bar{\Lambda}$. For Class II and Class III patterns, $\sigma^2(R)$ diverges and $\bar{\Lambda}$ is not finite. Appropriate generalized metrics are:

$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R} \quad \text{(Class II)}, \qquad A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}} \quad \text{(Class III)}. \tag{24}$$

For the period-doubling chain we measure $C_{II} = 0.123 \pm 0.016$ from the plateau of $\sigma^2(R)/\ln R$ at large R. For the 0222 chain we obtain $A_{III} = 0.151 \pm 0.014$ using the exact $\alpha = 0.639$ from the eigenvalue formula (see Fig. 10 in the Appendix). These metrics provide characterizations for Class II and Class III patterns analogous to $\bar{\Lambda}$ within Class I, enabling comparison of the magnitude of fluctuations across all three universality classes on appropriate scales.

6 Complete Catalog and Ranking

We have characterized 27 one-dimensional point patterns spanning all three hyperuniformity classes. Table 2 presents selected representative results, organized by class and sorted by $\bar{\Lambda}$ within Class I.

Table 2: Selected hyperuniformity results from the 27 point patterns analyzed in this work. Class I patterns are ranked by $\bar{\Lambda}$; the perfect integer lattice ($\bar{\Lambda} = 1/6$) provides the theoretical minimum. Stealthy patterns have $S(k) = 0$ over a finite exclusion zone (not a power law), parameterized by K^*; the exponent α is not applicable. Classes II and III have divergent $\bar{\Lambda}$; the generalized metrics C_{II} and A_{III} (Eq. 24) are reported instead.

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	0.250
Cubic α=2 (new)	*I*	*2.0*	*0.275*
Bronze	I	3.0	0.282
Copper	I	3.0	0.293
Nickel	I	3.0	0.310
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	—†	0.357
Bombieri-Taylor	**I**	**1.545**	**0.377**
Stealthy ($\chi = 0.2$)	I	—†	0.526
Stealthy ($\chi = 0.1$)	I	—†	1.021
Period-Doubling	II	1.0	div. ($C_{II} = 0.123$)
0222 Chain	III	0.64	div. ($A_{III} = 0.151$)

† Stealthy patterns: $S(k) = 0$ by construction for $|k| < K^*$; α is not a meaningful power-law exponent. The correct descriptor is $K^* = 2\pi\rho\chi$.

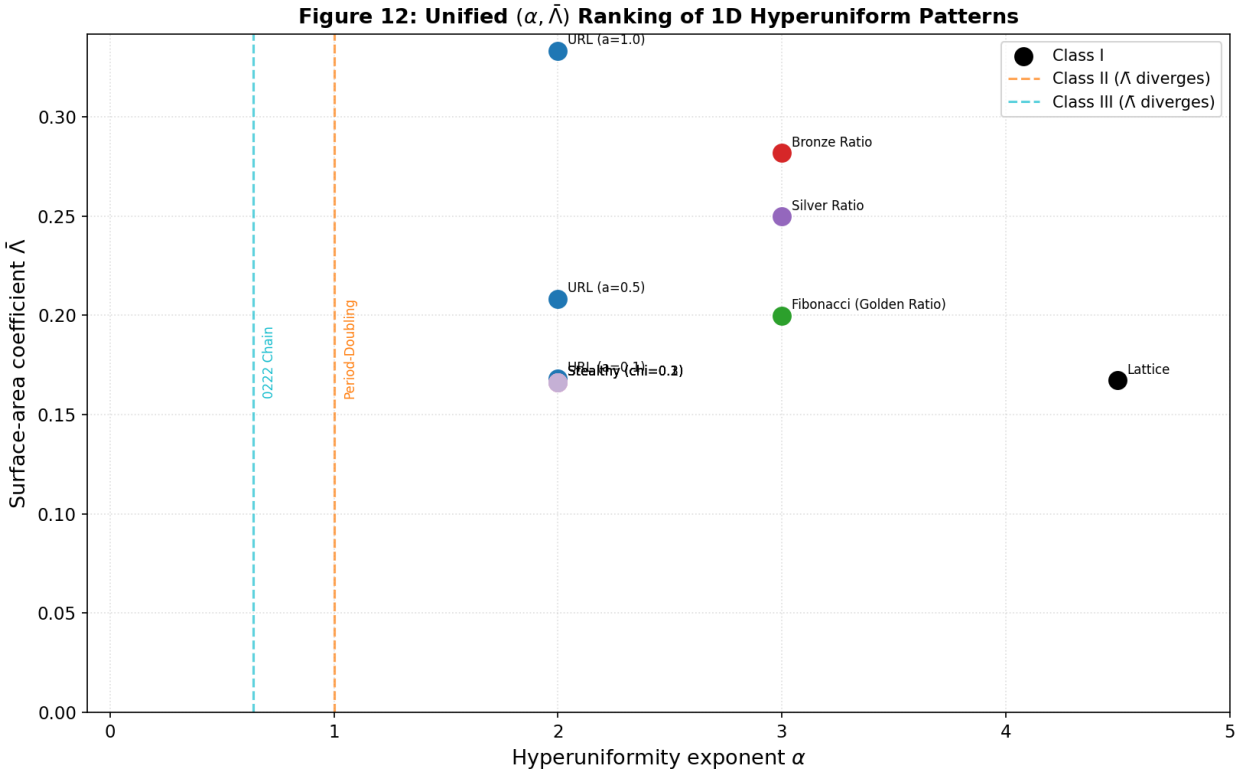

Figure 6: Complete ranking of hyperuniform patterns by $\bar{\Lambda}$ within Class I, and by α across classes. The Bombieri-Taylor cubic (highlighted) fills the gap between the URL/stealthy cluster ($\alpha = 2$) and the Period-Doubling chain ($\alpha = 1$). The horizontal dashed line marks the perfect-lattice minimum $\bar{\Lambda} = 1/6 \approx 0.167$.

Several observations emerge from Table 2 and Fig. 6. First, the perfect lattice achieves the theoretical minimum $\bar{\Lambda} = 1/6$: any deviation from perfect periodicity, no matter how small, increases $\bar{\Lambda}$. Second, α and $\bar{\Lambda}$ are genuinely independent metrics: the URL at $a = 0.1$ ($\alpha = 2$, $\bar{\Lambda} = 0.168$) is nearly as ordered as the perfect lattice in terms of $\bar{\Lambda}$, while the Fibonacci chain ($\alpha = 3$, $\bar{\Lambda} = 0.201$) is more ordered by the α metric but less so by $\bar{\Lambda}$. Third, the stealthy patterns at $\chi = 0.1$ and $\chi = 0.2$ have the largest $\bar{\Lambda}$ values in Class I; their fluctuations at short wavelengths are comparatively large even though long-wavelength fluctuations are perfectly suppressed over the entire exclusion zone. Fourth, the cubic α=2 tiling introduced in Section 5.6 occupies the same α value as the URL but with $\bar{\Lambda} = 0.275$, intermediate between the Silver chain ($\bar{\Lambda} = 0.250$) and the Bronze chain ($\bar{\Lambda} = 0.282$); it demonstrates that deterministic quasiperiodic patterns can share an α value with stochastic constructions. Finally, the Bombieri-Taylor cubic remains the sole occupant of the $1 < \alpha < 2$ interval, bridging the gap between two distinct physical mechanisms (URL/stealthy-type $\alpha = 2$ and boundary $\alpha = 1$ from period-doubling).

We also verified numerically that $\bar{\Lambda}$ is invariant under uniform density rescaling in one dimension: multiplying all point positions by a factor ρ (so that the density changes from ρ to 1) leaves $\bar{\Lambda}$ unchanged. This confirms that direct comparison of $\bar{\Lambda}$ values across patterns with different intrinsic densities — the metallic-mean chains all have different ρ — is valid without any normalization. Finite-size convergence of all variance-based metrics as a function of N is shown in Fig. 11 in the Appendix.

7 Conclusion

We have carried out a systematic numerical study of hyperuniformity in one-dimensional point patterns, analyzing 27 systems spanning all three universality classes and a wide range of hyperuniformity exponents $0.64 \leq \alpha \leq \infty$. Several conclusions follow from this work.

First, all five 2×2 metallic-mean substitution chains (Fibonacci, Silver, Bronze, Copper, and Nickel) yield $\alpha = 3$ by a determinant constraint on their integer substitution matrices. This result, which we derive analytically and confirm numerically to within 2% for each chain, establishes a fundamental limitation of the 2×2 family: it cannot access any Class I exponent other than $\alpha = 3$ without breaking the metallic-mean form.

Second, Class II ($\alpha = 1$) and Class III ($\alpha \approx 0.64$) behavior is demonstrated explicitly for the period-doubling and 0222 chains, respectively, with variance growth laws in quantitative agreement with theoretical predictions. Generalized scaling metrics C_{II} and A_{III} (Eq. 24) are introduced to characterize these patterns on a common footing with the Class I surface-area coefficient $\bar{\Lambda}$.

Third, the Bombieri-Taylor cubic substitution fills the gap $1 < \alpha < 2$. By moving from a 2×2 to a 3×3 substitution matrix, one breaks the eigenvalue inverse-product constraint and gains access to intermediate exponents. The measured $\alpha = 1.545$ agrees with Eq. (20) to within numerical uncertainty and the saturated number variance confirms Class I status.

Fourth, an exhaustive search of 39,304 3×3 non-negative integer matrices with row sums ≤ 4 (of which 4,882 satisfy the Pisot condition) proves that the interval $2 < \alpha < 3$ *cannot* be accessed at algebraic degree three: all unimodular matrices with complex conjugate eigenvalue pairs yield $\alpha = 2$ exactly (Eq. 22), and no all-real case in the search falls in $(2, 3)$. The search does, however, yield *to our knowledge the first deterministic substitution tiling with* $\alpha = 2$ — the cubic α=2 chain with $\bar{\Lambda} = 0.275$ — which provides a quasiperiodic structural analogue of the stochastic URL.

Fifth, the stealthy hyperuniform patterns analyzed here exhibit two distinct phases depending on initial conditions: an ordered-stealthy phase (arising from lattice-based initialization, $\bar{\Lambda} \approx 1/6$) and a disordered-stealthy phase (from the collaborating ensemble with random initialization, $\bar{\Lambda} \approx 1/(\pi^2\chi)$). Both satisfy the defining constraint $S(k) = 0$ for $|k| < K^*$, but differ substantially in their short-range order. This confirms that stealthy hyperuniformity does not uniquely determine a pattern; the χ and K^* descriptors are more appropriate than a power-law exponent α for characterizing these systems.

Open questions remain. Accessing $\alpha \in (2, 3)$ deterministically requires 4×4 or higher-rank substitution matrices; designing such systems with controlled α is a concrete next step. The relationship between the algebraic properties of the characteristic polynomial (Pisot versus Salem numbers) and the hyperuniformity class deserves further exploration. The Bombieri-Taylor chain has only a single representative in the $1 < \alpha < 2$ range; systematic enumeration of 3×3 Pisot matrices with three real eigenvalues may reveal a continuous family here as well. More broadly, extending the analysis to two-dimensional substitution tilings — Penrose, Ammann-Beenker, and related patterns — would clarify whether the 1D eigenvalue formula Eq. (8) generalizes to higher dimensions with the same predictive power.

Acknowledgements

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a collaborating group member; we are grateful for access to these data. This project was carried out as part of the Princeton Junior Independent Work program.

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A Additional Figures and Derivations

This appendix collects supplementary figures and derivations referenced in the main text.

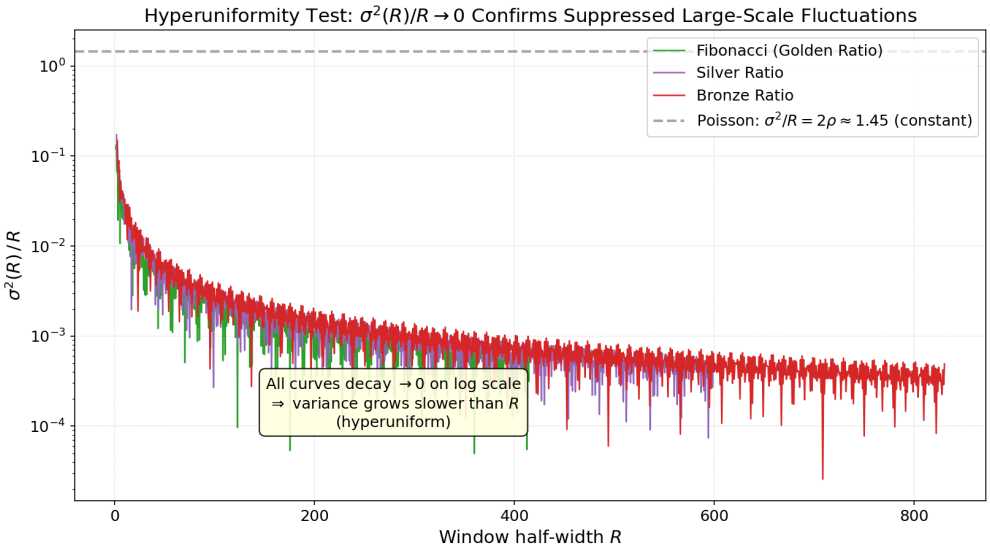

Figure 3: For random (Poisson) patterns, $\sigma^2/R = 2\rho = \text{const}$. For hyperuniform patterns, this ratio vanishes.

Figure 7: Hyperuniformity test plot showing $\sigma^2(R)/(2\rho R)$ versus R for selected patterns. Curves that approach zero confirm hyperuniformity; the horizontal Poisson baseline ($= 1$) serves as reference. Class I patterns approach zero as $\sim 1/R$, Class II as $\sim \ln R/R$, and Class III as $\sim R^{-\alpha}$.

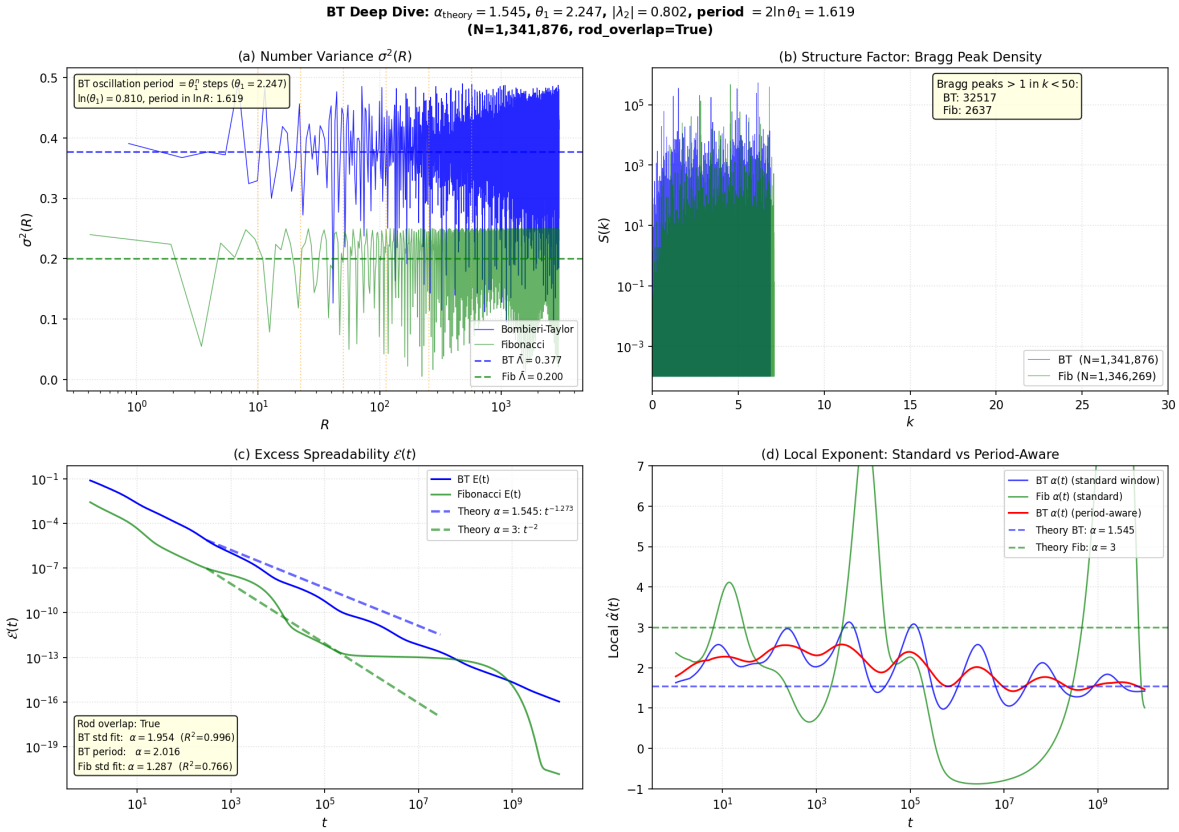

Figure 8: Bombieri-Taylor deep analysis. *(a)* Fine-resolution number variance $\sigma^2(R)$ showing the strong log-periodic oscillations with period $2\ln\theta_1 \approx 1.62$ in $\ln R$. *(b)* Structure factor $S(k)$ over a wide k-range, revealing the dense Bragg peak spectrum (32,517 peaks in $k < 50$, compared to 2,637 for Fibonacci). *(c,d)* Spreadability $\mathcal{S}(t)$ with standard and period-aware power-law fits. The large $|\lambda_2| = 0.802$ causes slow convergence; at $t \in [10^5, 10^9]$ the fit yields $\hat{\alpha} = 1.67$, still above the theoretical value 1.545.

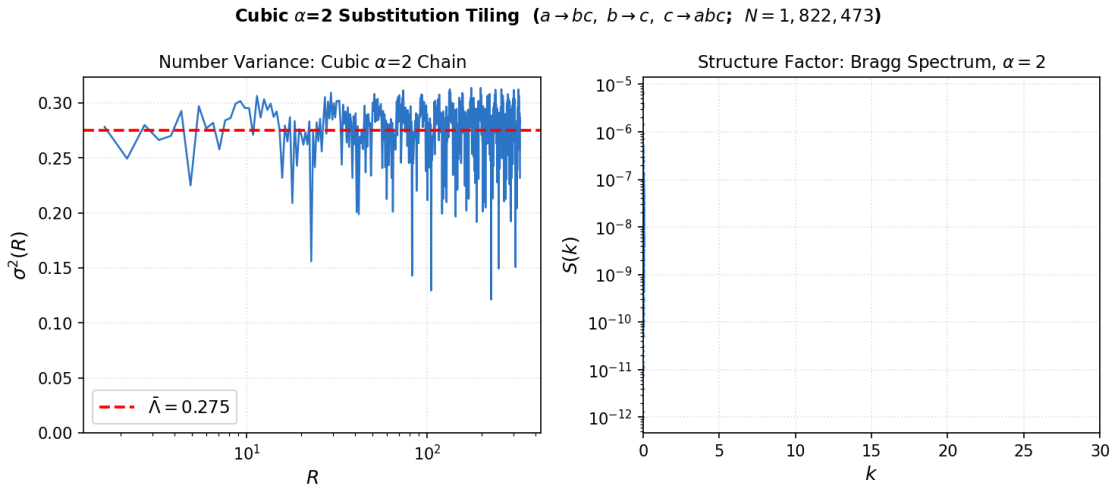

Figure 9: Hyperuniformity analysis of the cubic α=2 substitution chain ($c \to abc$, $a \to bc$, $b \to c$; $N \approx 1.8 \times 10^6$). *(Left)* Number variance $\sigma^2(R)$ is bounded (Class I), saturating at $\bar{\Lambda} = 0.275$. *(Right)* Structure factor shows the expected Bragg-peak spectrum, consistent with $\alpha = 2$ from Eq. (22).

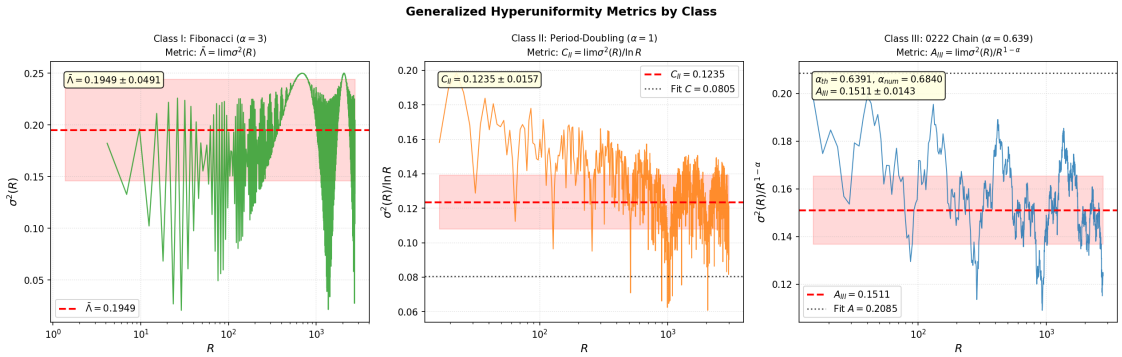

Figure 10: Generalized scaling metrics for Class II and Class III patterns. *(Left)* Period-Doubling: normalized variance $\sigma^2(R)/\ln R$ approaches a constant $C_{II} = 0.123 \pm 0.016$ at large R, confirming the logarithmic scaling law. *(Right)* 0222 chain: $\sigma^2(R)/R^{1-\alpha}$ (with $\alpha = 0.639$ from the eigenvalue formula) approaches $A_{III} = 0.151 \pm 0.014$, confirming the power-law scaling.

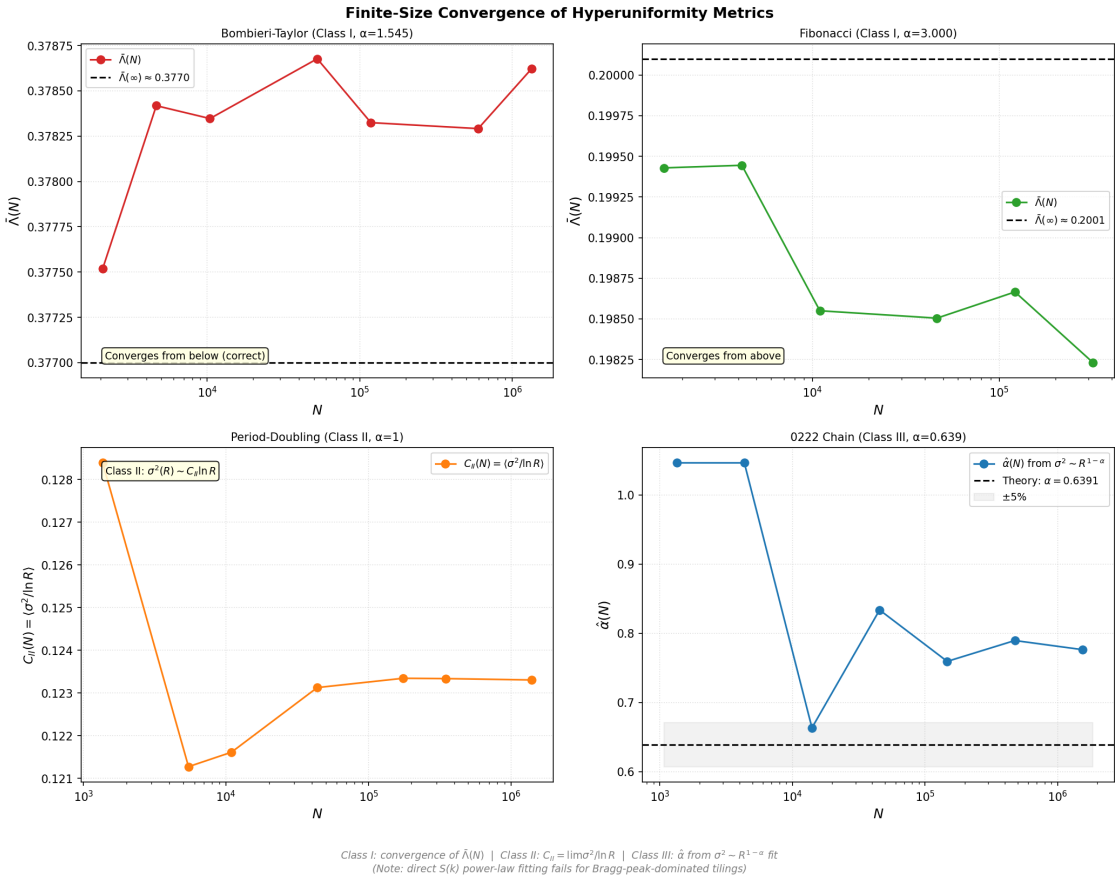

Figure 11: Finite-size convergence of variance-based hyperuniformity metrics as a function of system size N. Class I metrics ($\bar{\Lambda}$ for Bombieri-Taylor and Fibonacci) converge rapidly and are stable for $N \gtrsim 2000$. The Class II metric C_{II} for the period-doubling chain requires $N \gtrsim 10^5$ for reliable estimation. The Class III exponent $\hat{\alpha}$ for the 0222 chain shows slow convergence due to $|\lambda_2| > 1$.